Hierarchical Shape, Morphometric Solvation, and SE(3)-Invariant Chemistry Residual Energy

$$E_{\text{total}}(\{T_A\}) = E_{\text{shape}}(\text{hard vdW}) + E_{\text{morph}}(V, A, C, X) + E^{(2B)}_{\text{chem}}(\text{HB},\ \text{WB},\ \text{Cl}^-,\ \text{desolv},\ \text{cosine}) + E^{(3B)}_{\text{junc}}(B_1\ \text{core},\ B_2\ \text{collar})$$

a Three-Tier Physical Decomposition on Monomer Heavy-Atom Geometry (1001 Atoms)

1. Hard-Core Steric & Morphometric Solvation Baseline

- Hard vdW Exclusion: $E_{\text{shape}} = +\infty$ if $d_{ij} < r_i + r_j$ ($g_{ij} < 0$)
- Hadwiger WBII Morphometry: $\Omega_{\text{wb2}} = -pV + \sigma A + \kappa C + \bar{\kappa} X$ ($r_s = 1.4$ Å, $\eta = 0.383$)
- Typed Gaussian Burial: $E_{\text{solv}} = g_{\text{burial}}(V_{\text{ov}}) \sum_i \tilde{w}_i \Delta V_i^{\text{buried}}$ (bounded at $\pm 12.5\, k_B T$)

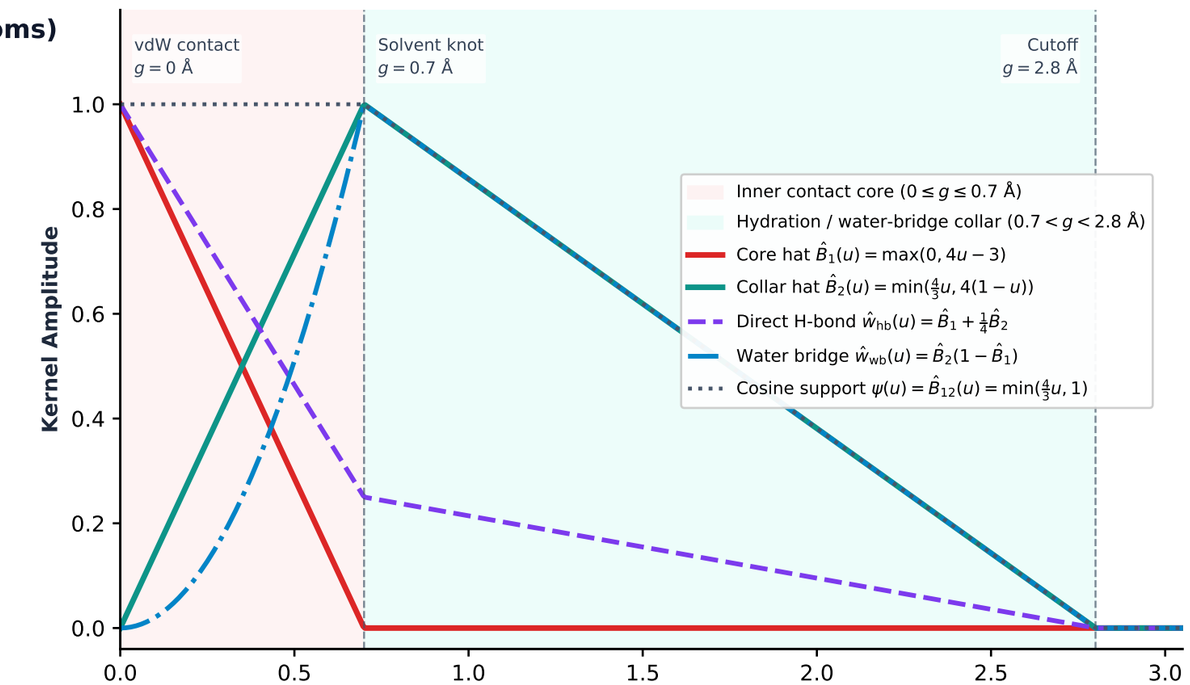
2. Two-Body Polar, Water-Bridge, Anion & Contrastive Cosine Residual

- 1-to-1 H-Bonds & Salt Bridges: $-w_{\text{hb}} S_{\text{hb}} - w_{\text{chb}} S_{\text{chb}} - w_{\text{Cl}} S_{\text{Cl}^-}$ ($g_{ij} < 2.8$ Å)
- Per-Atom Saturated Water Bridge: $S_{\text{wb}} = \frac{1}{2} \sum_i \hat{w}_{\text{wb}}(u^{\text{comp}}_{i,\max})\, e^{-1.5 S_{\text{clash}}}$ (no wb_cap)
- Unpaired Burial & Clash Penalty: $+w_{\text{clash}} S_{\text{clash}} + w_{\text{unp,pol}} U_{\text{pol}} + w_{\text{unp,chg}} U_{\text{chg}}$
- Contrastive Cosine Match: $E_{\cos} = -w^{\pm}_{\cos} S^{AB}_{\cos}$, $s_{ij} = \frac{1}{2}(\mathbf{u}_i \cdot \mathbf{v}_j + \mathbf{v}_i \cdot \mathbf{u}_j) \in [-1, 1]$

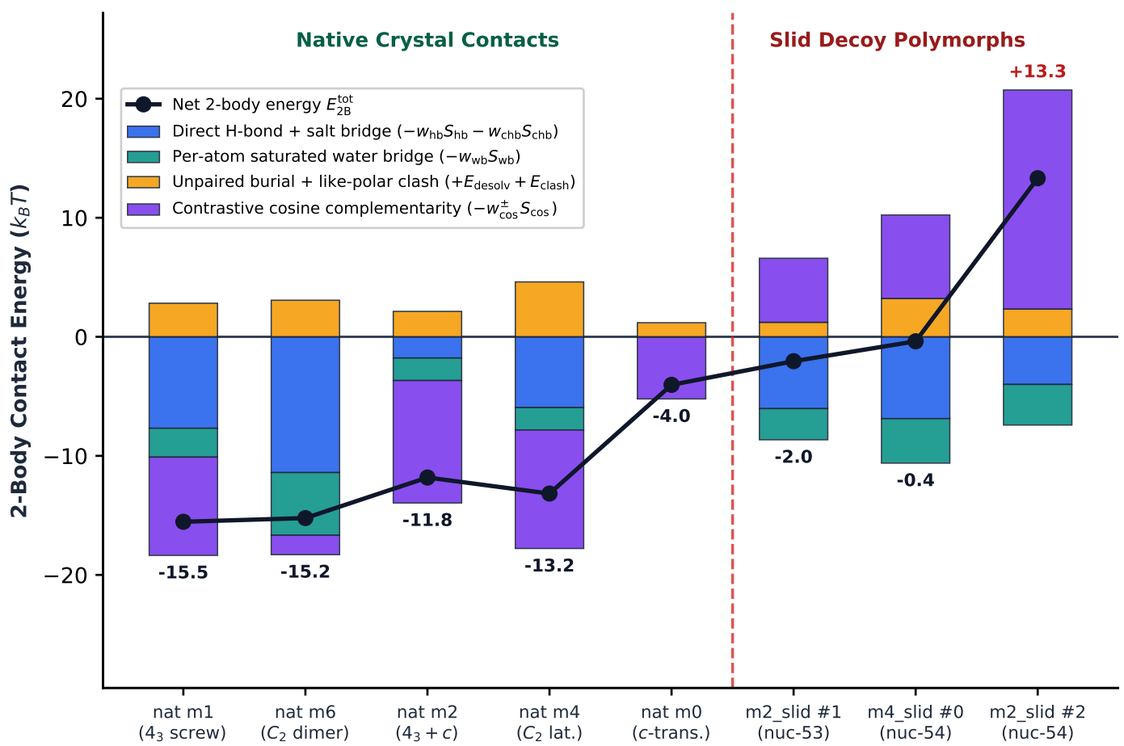
3. Cutoff-Free Three-Body Vertex Surface-Field Hamiltonian (Core + Collar)

- 11D Surface Feature Field on A: $\boldsymbol{\psi}_i = \mathbf{n}_i \oplus \gamma_c a_i (\mathbf{u}_i - \mathbf{v}_i) \in \mathbb{R}^{11}$
- Inner-Core Steric Crowding (B1): $E^A_{\text{crowd}} = +w_{\text{sat}} \sum_{B<C} \Big\| \sum_{i \in A} B_1(u^B_i) B_1(u^C_i) \boldsymbol{\psi}_i \Big\|$
- Hydration-Collar Seam Sealing (B2): $E^A_{\text{seal}} = -w_{\text{seal}} \sum_{B<C} g_{AB} g_{AC} \frac{\|\mathbf{V}^A_{\text{col},BC}\|^2}{\|\mathbf{V}^A_{\text{col},BC}\| + w_0}$
- Desolvation Sharing (Ci): $E^A_{\text{coop}} = -\sum_{i \in A} w_{\text{coop}}(f_i) \max(0,\, S^{\text{tot}}_{i,\text{comp}} - S^{\max}_{i,\text{comp}})$

b Cutoff-Free Radial Basis Functions on Gap $g_{ij} = d_{ij} - (r_i + r_j)$



c Exact 2-Body Energy Decomposition on Native & Slid Poses ($k_B T$)



d Basin Association Constants $\ln(K/\text{Å}^3)$ from `pair-binding`

